

MLOps Assignment 2:

Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment

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1. Introduction & Objective

The objective of this assignment was to build a complete MLOps pipeline by fine-tuning a pre-trained language model on the UCSD Goodreads book review dataset for genre classification. The workflow included loading a pre-trained model from Hugging Face, training on Kaggle GPU, tracking experiments, evaluating the model, and deploying it to the Hugging Face Hub.

This assignment helped in understanding production-grade ML workflows using modern tools like Hugging Face Transformers, Kaggle Notebooks, and Weights & Biases.

2. Model Selection Rationale

I selected **DistilBERT (distilbert-base-uncased)** for this task.

DistilBERT is a distilled version of the original BERT model. According to its official model card on Hugging Face, it retains **97% of BERT's language understanding capabilities** while being **40% smaller and 60% faster**. This makes it highly suitable for fine-tuning on Kaggle's free GPU resources, especially with limited GPU hours.

Why DistilBERT was a good choice:

- Smaller model size → Lower memory usage
- Faster training and inference
- Strong performance on text classification tasks
- Excellent contextual understanding, which helps in distinguishing subtle differences between book genres like "Fantasy & Paranormal" vs "Romance"

The model was loaded with 9 output labels corresponding to the book genres in the dataset.

3. Kaggle Training Setup

Environment Configuration:

- **Platform:** Kaggle Notebook
- **Accelerator:** GPU T4 × 2
- **Internet:** Enabled
- **Secrets Used:** WANDB_API_KEY and HF_TOKEN

Dataset Preparation:

- Used a sampled version of the Goodreads dataset (~400 records, 4 genres in final sample due to fallback data loading).
- Performed train-test split (80-20).
- Tokenized using DistilBERT tokenizer with max_length=512.

Training Configuration:

- Model: distilbert-base-uncased
- Epochs: 3
- Batch Size: 16
- Learning Rate: 3e-5 (default)
- Mixed Precision (fp16=True)

Due to persistent W&B authentication issues in the Kaggle environment, training was completed with report_to="none".

Final Training Results:

- **Accuracy:** 0.2625
- **F1 Score:** 0.2249
- **Eval Loss:** 3.2795

4. Model Evaluation

Final evaluation was performed on the test set using the Hugging Face Trainer.

Key Metrics:

	Metric	Score
Eval Loss		3.2795
Accuracy		0.2625
F1 Score		0.2249

A detailed classification report was generated and saved as eval_report.json. Some precision warnings appeared due to the small dataset and class imbalance, which is expected.

5. Model Deployment to Hugging Face Hub

The trained model and tokenizer were successfully pushed to the Hugging Face Hub.

Model Link: <https://huggingface.co/BhavikSolanki/ML-OPS-Assignment2>

This allows anyone to load the model using:

Python

```
from transformers import pipeline
pipe = pipeline("text-classification", model="BhavikSolanki/ML-OPS-Assignment2")
```

6. Challenges & Learnings

Challenges Faced:

1. Dataset loading issues — had to use fallback sample data as external URLs were not working reliably.
2. Persistent W&B authentication errors in Kaggle (JSON parsing issue).
3. Small dataset size leading to relatively low accuracy.
4. Managing secrets correctly in Kaggle (initially named secret incorrectly).

Learnings:

- Understood the complete MLOps lifecycle: data prep → training → evaluation → deployment.
- Learned how to use Kaggle Secrets for secure credential handling.
- Gained hands-on experience with Hugging Face Hub for model sharing.
- Realized the importance of modular code and reproducible pipelines.
- Understood practical constraints of free-tier GPU platforms.

What I Would Do Differently:

- Use a larger dataset by properly uploading it to Kaggle Input.
- Try using a different model like bert-base-uncased if time permits.
- Implement proper W&B tracking using offline mode + manual sync.

7. Conclusion

This assignment successfully demonstrated a full MLOps pipeline from training on Kaggle to deploying the model on Hugging Face Hub. While the accuracy is modest due to limited data, the focus was on understanding the workflow rather than achieving state-of-the-art performance.

Submission Links:

- **GitHub Repository:** <https://github.com/bhavik-solanki-G25AIT2106/ML-OPS-Assignment2>
- **Kaggle Notebook:** <https://www.kaggle.com/code/g25ait2106/ml-ops-assignment-2>
- **Hugging Face Model:** <https://huggingface.co/BhavikSolanki/ML-OPS-Assignment2>
- **W&B Project:** <https://wandb.ai/bhaviksolanki-indian-institute-of-technology-jodhpur/mlops-assignment2?nw=nwuserbhaviksolanki>